

Ionospheric Scintillation Effects across Multiple Carrier Frequency Bands Transmitted from LEO Satellites

Andrew K. Sun, *Korea Advanced Institute of Science and Technology*

Y. Jade Morton, *University of Colorado Boulder*

Jiyun Lee, *Korea Advanced Institute of Science and Technology*

Biographies

Andrew Kiyoun Sun is a Ph.D. student in Aerospace Engineering at the Korea Advanced Institute of Science and Technology (KAIST) in South Korea, under the supervision of Dr. Jiyun Lee. He is a visiting Ph.D. student in Dr. Jade Morton's Satellite Navigation and Sensing Lab at the University of Colorado Boulder from April to October 2023 while working on the project presented in this paper. His research interests encompass the study of dynamics in ionospheric anomalies and their impact on GNSS applications.

Dr. Y. Jade Morton is Helen and Hubert Croft Professor in the Ann and H.J. Smead Aerospace Engineering Sciences Department at University of Colorado Boulder. Her research interests lie at the intersection of satellite navigation technologies and remote sensing of the Earth's space environment, atmosphere, and surface. She served as a President of the ION and is a member of the US Space-based PNT Advisory Board. She received an AGU SPARC award, ION Burka, Thurlow, and Kepler awards, and IEEE PLANS Richard Kershner award. Dr. Morton received her Ph.D. in EE from Penn State. She is a fellow of IEEE, ION, and UK's Royal Institute of Navigation.

Dr. Jiyun Lee is a KAIST Endowed Chair Professor in the department of Aerospace Engineering at the Korea Advanced Institute of Science and Technology (KAIST). Her specialties lie in the area of GNSS augmentation systems, integrity architecture for multi-sensor integrated navigation systems, and GNSS remote sensing. She received her Ph.D. from Stanford University in aeronautics and astronautics. She served ION as secretary of Satellite division executive committee (2022-2024), technical representative of ION council (2019-2021), and pacific technical advisor (2016-2018, 2020-2022).

Abstract

Prior simulation studies have indicated that signals from LEO satellites exhibit higher signal dynamics and more significant scintillation effects compared to those transmitted from MEO satellites. This study employs a physics-based simulation model initiated with real GPS scintillation data. It extends the analysis to assess scintillation effects transmitted from LEO satellites across various carrier frequencies in the L band (L1, L2, and L5) and other bands (VHF, UHF, and S). The results provide quantitative evidence that signals at lower frequencies experience more frequent fades and higher phase dynamics. Analyses across various frequency bands validate that distinct signal dynamics determine the time scale of scintillation, resulting in temporal characteristics of frequency-dependent signal fading. The simulation also demonstrates a consistent rate of selective fading among different L-band signals across different signal dynamics. This frequency-selective fading allows the adaptation of inter-frequency aiding techniques in receivers to improve the robustness of LEO signal processing.

1 INTRODUCTION

Ionospheric scintillation refers to the rapid fluctuations in radio wave amplitude and phase as the signal traverse ionospheric plasma irregularities. It is well-known that these fluctuations have a significant impact on space-based applications, including the Global Navigation Satellite System (GNSS), which transmits L-band radio signals to receivers on or near the surface of the Earth. In severe

cases, ionospheric scintillation can lead to deep signal fades and abrupt phase changes, disrupting the receiver's carrier tracking process and potentially degrading GNSS navigation performance.

Recent simulation studies indicated that scintillation effects pose greater challenges for signals transmitted from low Earth orbit (LEO) satellites compared to GNSS signals from medium Earth orbit (MEO) satellites. Simulation studies have been conducted to analyze the impact of scintillation on L1 C/A signals propagating from LEO satellites to ground receivers (Morton et al., 2022) and assessed their signal tracking performance (Xu et al., 2023). LEO satellite signals exhibit higher variability in signal dynamics, resulting in more frequent signal fades with higher phase dynamics during scintillation, under identical irregularity conditions (Morton et al., 2022). This leads to more degraded performance of signal tracking with signals transmitted from LEO satellites than MEO satellites (Xu et al., 2023). Potential applications utilizing signals from LEO satellites have gained increasing attention for future navigation applications (Kassas, 2020; Prol et al., 2022; Reid et al., 2020), or augmenting GNSS services (Enge et al., 2012; Joerger et al., 2010), and supporting remote sensing applications (Garrison et al., 2017). Therefore, understanding scintillation effects on signals transmitted from LEO satellites is crucial to ensure these applications.

This work extends prior simulation-based study of the scintillation effects on a single L-band frequency signal transmitted from LEO satellite to analyzing these effects across diverse carrier frequencies. Commercial LEO satellites transmit signals at various frequency bands, including lower bands like very high frequency (VHF), ultra-high frequency (UHF) and higher bands like S-band, as well as the L-band, providing a diverse set of challenges and opportunities for navigation and communication systems. Investigating scintillations across these signals at various frequency bands provides valuable insights into potential scintillation impacts, contributing to the development of robust and reliable LEO satellites-based/augmented applications, particularly in the face of such ionospheric threats. For signals that are in nearby frequency bands, multi-frequency signals from LEO satellites hold particular interests due to their potential applications leveraging frequency diversity.

The lack of real scintillation data collected from LEO satellite transmissions presents a challenge in understanding scintillation effects on signals at various frequency bands. To address this challenge, a physics-based scintillation simulation model becomes a practical approach for investigating scintillation signals experiencing various satellite-to-receiver platform dynamics. A scintillation simulation tool developed in Rino et al. (2018) and updated in Jiao et al. (2018) is based on the phase screen theory of radio wave propagation through ionospheric structures. This model generates scintillation realizations through phase screen calculations, providing statistical representations of ionospheric irregularities specified by five spectral parameters of the scintillation signal. The phase screen model incorporates space-to-time conversion using a scale factor ρ_F/v_{eff} , where ρ_F is the spatial scale of irregularities and v_{eff} is the radio wave effective scan velocity across ionosphere. Xu et al. (2020) developed a compact version of this model by reducing the number of input parameters from five to two: the amplitude scintillation index S_4 and the scintillation signal decorrelation time τ_0 . This model provides a statistically consistent abstraction of ionospheric irregularities using approximated phase-changing screens, enabling simulations of scintillations with different τ_0 values associated with various signal dynamics.

This study employs the same physics-based, data-consistent, compact scintillation simulator (Xu et al. 2020) to investigate the impact of ionospheric irregularities on signals transmitted from LEO satellites to ground receivers across various carrier frequencies. In Section 2, we introduce the scintillation simulator initialized with real scintillation data, enabling us to simulate scintillation across these different frequencies. Section 3 presents simulation scenarios of signal dynamics for various signal transmission configurations and carrier frequencies. Section 4 provides the scintillation characteristics across the different frequencies and signal dynamics. The resulting signal fades and phase changes are analyzed to investigate the scintillation effects on signals transmitted from LEO satellites across diverse frequency bands and to compare these findings with signals from MEO satellites. Conclusions and future works are presented in Section 5.

2 SCINTILLATION SIMULATOR FOR DIFFERENT FREQUENCIES

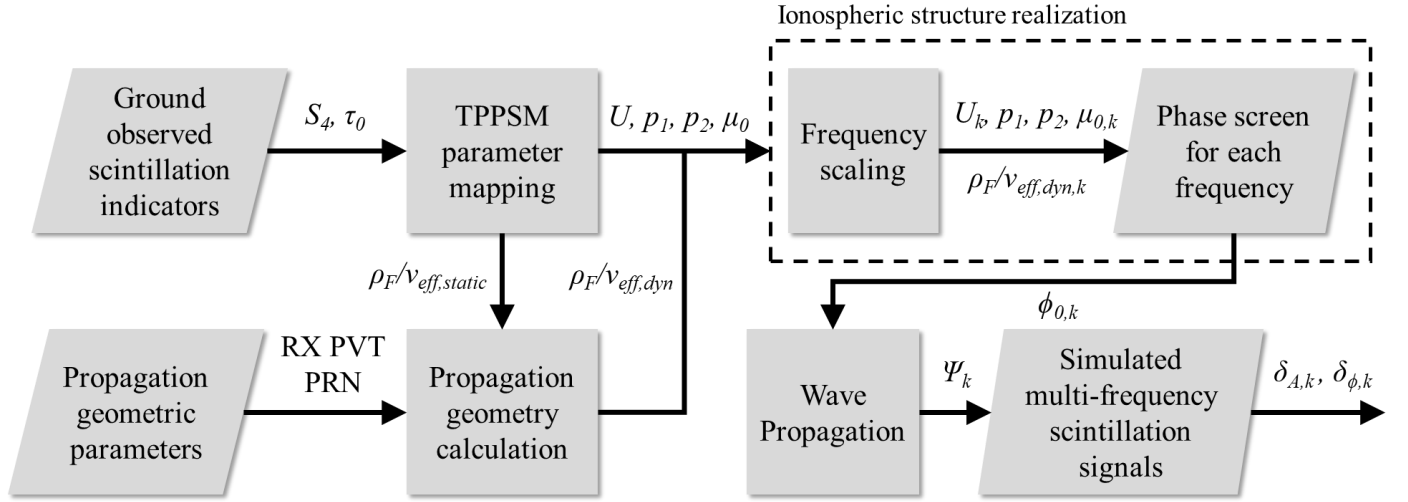


FIGURE 1 Flow chart of multi-frequency scintillation simulation based on the simulator presented in Xu et al. (2020).

The ionospheric scintillation simulator used in this study is based on the model published in Xu et al. (2020). Figure 1 illustrates the flow chart of the multi-frequency scintillation simulation using the model introduced in Xu et al. (2020). This model relies on only two input parameters: S_4 , the amplitude scintillation index characterizing the magnitude of the amplitude fluctuation, and τ_0 , the decorrelation time capturing the temporal variability of signal fluctuations. These parameters are derived from ground-based GNSS measurements of scintillation signal, serving as initiation data for the simulation.

These two parameters can be translated into a set of five parameters for a two-component power-law phase screen model (TPPSM): $\{U, \rho_F/v_{\text{eff}}, p_1, p_2, \mu_0\}$, where U is the universal scattering strength, ρ_F/v_{eff} is the space-to-time scaling parameter, p_1 and p_2 are the power-law indices, and μ_0 is the spectral break wavenumber (Jiao et al., 2018; Rino et al., 2018). This mapping is established based on the numerical relationship between $\{S_4, \tau_0\}$ and $\{U, \rho_F/v_{\text{eff}}\}$, while assuming typical values for the spectral parameters $\{p_1, p_2, \mu_0\}$, as derived from real data analysis (Xu et al., 2020).

As previously presented in Morton et al. (2022), the simulator accommodates various scenarios related to signal propagation geometry. These scenarios are simulated using distinct $\rho_F/v_{\text{eff,dyn}}$ values designed to align with the user-defined satellite-receiver geometry. The calculation of $\rho_F/v_{\text{eff,dyn}}$ involves considering the signal scan velocity at the phase screen and the drift velocity of the irregularities, v_{drift} . The unknown v_{drift} is determined through numerical solutions by comparing the $\rho_F/v_{\text{eff,static}}$ values derived from satellite orbits and those inferred from τ_0 observations during scintillation events. The estimated v_{drift} is then incorporated into the calculation of $\rho_F/v_{\text{eff,dyn}}$ for the desired signal dynamics, considering the satellite-receiver geometry, resulting in the corresponding ρ_F values and the signal scan velocity at the phase screen (Xu et al., 2020). Further details regarding different scenarios of satellite-receiver geometry and signal dynamics will be provided in Section 3.

Among the TPPSM parameters, U , μ_0 , and ρ_F/v_{eff} depend on the signal carrier frequency, while p_1 and p_2 remain consistent across different frequencies. To establish phase screen model for different frequencies, these three parameters are scaled from a reference frequency (f_r) to another frequency (f_k) using the following equations (1)–(3) (Xu et al., 2020):

$$U(f_{L5}) = \begin{cases} U(f_r) \left(\frac{f_r}{f_k} \right)^{\frac{p_1+3}{2}} & \text{if } \mu_0(f_r) \geq 1, \mu_0(f_k) \geq 1 \\ U(f_r) \left\{ \mu_0(f_r) \right\}^{p_1-p_2} \left(\frac{f_r}{f_k} \right)^{\frac{p_1+3}{2}} & \text{if } \mu_0(f_r) < 1, \mu_0(f_k) \geq 1 \\ U(f_r) \left\{ \mu_0(f_k) \right\}^{p_2-p_1} \left(\frac{f_r}{f_k} \right)^{\frac{p_1+3}{2}} & \text{if } \mu_0(f_r) \geq 1, \mu_0(f_k) < 1 \\ U(f_r) \left(\frac{f_r}{f_k} \right)^{\frac{p_2+3}{2}} & \text{if } \mu_0(f_r) < 1, \mu_0(f_k) < 1 \end{cases} \quad (1)$$

$$\mu_0(f_k) = \mu_0(f_r) \sqrt{f_r / f_k} \quad (2)$$

$$\rho_F / v_{\text{eff}}(f_k) = \rho_F / v_{\text{eff}}(f_r) \sqrt{f_r / f_k} \quad (3)$$

Based on the established phase screen model, a plane wave is propagated through the phase screen realizations, with the user-specified signal propagation geometry and satellite-receiver platform dynamics embedded in $\rho_F / v_{\text{eff, dyn}}$. The same random number generator is used to generate phase screens at different frequencies, considering that signals across different frequencies propagate through the same ionospheric plasma structures. Consequently, this process generates simulated scintillation signal wave fields (Ψ_k) at the receiver. These, in turn, result in scintillation-induced amplitude ($\delta_{A,k}$) and carrier phase fluctuation time series ($\delta_{\phi,k}$) at a given frequency (f_k).

The scintillation simulation model was designed to handle L-band signals. However, some fundamental assumptions such as scintillation is only caused by forward scattering may be violated at low frequencies such as the VHF signal presented in this paper. There are on-going efforts to extend the model to the lower frequencies.

3 SIMULATION SCENARIOS

This study simulated scintillations using a 5-min segment of initiation data observed in GPS PRN24 satellite L1 CA signals collected by a ground station in Hong Kong on October 5th, 2013, from 12:25 to 12:30 UT (Morton et al., 2022). The Scintillation simulations utilized input parameters of S_4 at 0.8 and τ_0 at 0.76 s, which represent the average values within the initiation data segment.

3.1 Signal Dynamics Scenarios

Based on the initiation data representing the MEO satellite scenario with $\rho_F / v_{\text{eff, static}}$, we established phase screen models. Then, signals at L-band frequencies (L1, L2, and L5) and other frequency bands (VHF, UHF, and S-band) were generated and propagated through the phase screens. Simulations considered several signal dynamics scenarios correspond to MEO and LEO satellite transmissions.

The same S_4 value of 0.8 (equivalent to U of 1.47) for L1 frequency was applied across all scenarios. Each scenario was characterized by distinct $\rho_F / v_{\text{eff, dyn}}$ values for each pair of the LEO satellite transmission and phase screen configuration. LEO satellite orbits were configured by projecting the satellite orbit from the initiation data (GPS PRN24 satellite) at MEO altitude to desired altitudes (h_{TX}) of 550 km and 800 km. The signal scan dynamics for the phase screen realizations were specified at the desired altitudes (h_{PS}) of 300 km and 500 km.

Table 1 outlines the five simulation scenarios where scenario 0 simulates the GPS PRN24 satellite at MEO altitude, and scenarios 1-4 represent LEO transmission cases with different pairs of h_{TX} and h_{PS} . Table 1 provides the range of the space-to-time scale factor (ρ_F / v_{eff}) for each scenario, with the maximum and minimum values for LEO satellite transmission scenarios highlighted in red text. To analyze representative scenarios of signal dynamics, we selected three specific cases: MEO with ρ_F / v_{eff} of 1.18, LEO-min with ρ_F / v_{eff} of 0.47, and LEO-max with ρ_F / v_{eff} of 0.07 (Morton et al., 2022). Each case represents the minimum and maximum signal dynamics for LEO satellite transmission scenarios required as an input for the scintillation simulator.

TABLET 1

Simulation Scenarios for Satellite Transmission and Phase Screen Configuration as Presented in Morton et al. (2022).

Scenario	h_{PS} (km)	h_{TX} (km)	ρ_F / v_{eff} (s)
0	350	MEO	[1.17, 1.19]
1	300	800	[0.24, 0.47]
2	300	550	[0.16, 0.43]
3	500	800	[0.20, 0.30]
4	500	550	[0.07, 0.14]

3.2 Simulation across Carrier Frequency Bands

For the selected signal dynamics and corresponding phase screen models of L1 (1575.42 MHz) signals, we applied equations (1)-(3) to perform frequency scaling of phase screen parameters. This process yielded corresponding phase screen realizations for L2 (1227.60 MHz), L5 (1176.45 MHz), and various frequency bands, including VHF (137.5 MHz), UHF (965 MHz) and S-band (2200 MHz), as potential options for current or future commercial LEO satellite transmission signals. The frequency of 137.5 MHz is allocated for VHF use in NOAA meteorological satellites (NOAA, 2009). The frequencies 965 MHz in UHF and 2200 MHz in S-band are employed for the RF beacon signals transmitted by the COSMIC-2 satellite (Hsiao et al., 2022).

Table 2 presents an overview of the scaled parameters: U , μ_0 , ρ_F/v_{eff} , along with the estimated S_4 and τ_0 values for each frequency. The frequency scaling results reveal that U exhibits higher values at lower frequencies, resulting in stronger scintillations with elevated S_4 values, while saturation occurs around 1. ρ_F/v_{eff} tends to have higher values at lower frequencies, indicating reduced signal dynamics. However, the variations in ρ_F/v_{eff} are much smaller than those in U at different frequencies, resulting in a less significant contribution to simulation results.

TABLET 2

Scaled Phase Screen Parameters and estimated S_4 Values for Different Frequencies across Signal Dynamics Scenarios.

Frequency	U	μ_0	ρ_F/v_{eff}			S_4
			MEO	LEO-min	LEO-max	
S (2200 MHz)	0.48	0.51	1.00	0.40	0.06	0.48
L1 (1575.42 MHz)	1.47	0.60	1.18	0.47	0.07	0.80
L2 (1227.60 MHz)	3.38	0.68	1.34	0.53	0.08	1.01
L5 (1176.45 MHz)	3.90	0.69	1.37	0.54	0.08	1.03
UHF (965 MHz)	7.58	0.77	1.51	0.60	0.09	1.08
VHF (137.5 MHz)	2376.85	2.03	3.99	1.59	0.24	1.03

3.3 Scintillation Realizations

For each simulation configuration, we generated 1,000 minutes of scintillation realizations with a sampling interval of 1 ms to obtain statistically significant scintillation characteristics. Figure 2 presents examples of scintillation realizations under different signal dynamics scenarios: MEO, LEO-min, and LEO-max, and different frequencies: L1, L2, and L5. Figure 2a clearly shows that higher signal dynamics result in more frequent deep fades and larger phase variations, consistent with the earlier analysis in Morton et al. (2022). Figure 2b shows simulation results for L1, L2, and L5 frequencies, indicating that lower frequencies at L2 and L5 experience much deeper fades compared to L1, in accordance with the larger S_4 values at L2 and L5 as shown in Table 2. Nevertheless, they exhibit similar overall patterns in signal intensity and phase variations, as simulations are conducted for signals across frequencies propagating through the same sequence of phase screens.

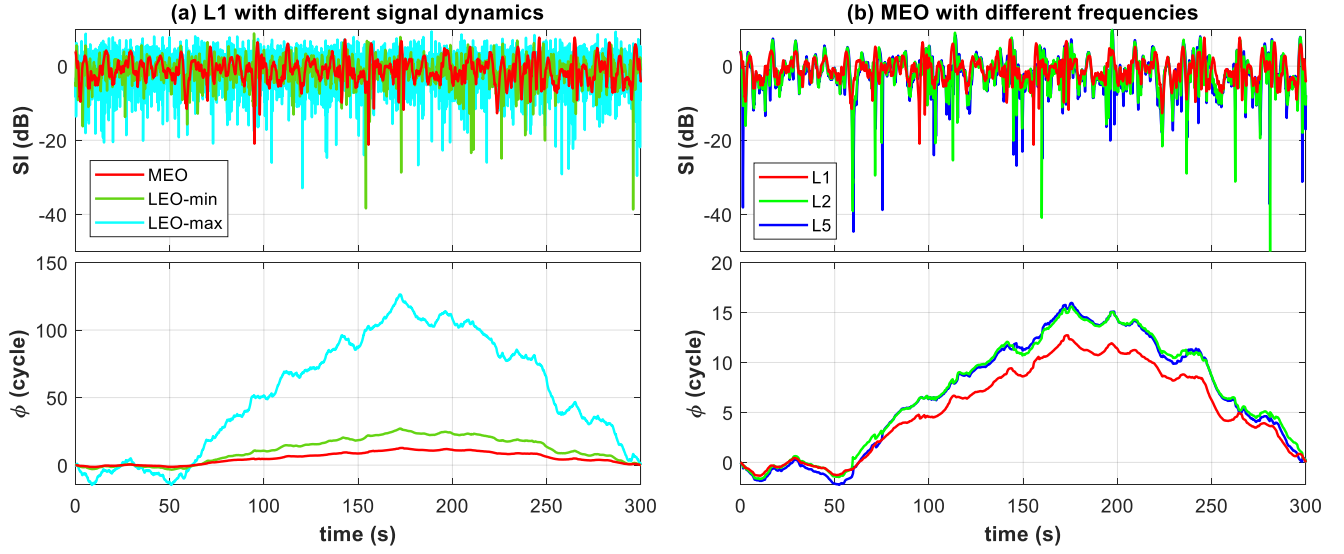


FIGURE 2 Simulated time series of scintillation signal intensity (SI) and carrier phase fluctuations (ϕ) over the 5-min simulation for (a) different signal dynamics at L1 frequency and (b) different frequencies at the MEO case.

Figure 3 shows the probability distribution functions (PDFs) and autocorrelation functions (ACFs) of signal intensity over 1,000-minute simulations. The results in Figure 3a indicate a consistent PDF of intensity across different signal dynamics despite their varying ρ_F/v_{eff} values. In the upper plot of Figure 3a, the intensity PDF appears nearly identical for all MEO, LEO-min, LEO-max cases as they share identical phase screen parameters resulting in a consistent distribution of signal intensity. However, the lower plot in Figure 3a shows the ACF of intensity, demonstrating a time scale proportional to ρ_F/v_{eff} . The decorrelation time, τ_0 , where the time lag at ACF reaches a value of e^{-1} , is 0.78 s, 0.31 s, and 0.05 s for ρ_F/v_{eff} of 1.18 s^{-1} , 0.48 s^{-1} , and 0.07 s^{-1} for MEO, LEO-min, and LEO-max, respectively. This proportional relationship between τ_0 and ρ_F/v_{eff} has been previously discussed in Xu et al. (2020).

On the other hand, the upper plot in Figure 3b reveals varying intensity PDFs across different frequencies, with lower frequencies exhibiting a higher probability distribution at lower signal intensity. This behavior is attributed to both frequency-scaled U and ρ_F/v_{eff} , as presented in Table 2. In the lower plot of Figure 3b, a similar time scale of signal intensity and τ_0 is observed, with values of 0.78 s, 0.72 s, 0.71 s for L1, L2, and L5, respectively. It is noteworthy that these values may not accurately exhibit the proportional relationship with the frequency-scaled ρ_F/v_{eff} , as U also varies with frequency, as discussed in Xu et al. (2020). Furthermore, under severe scintillations, the saturation of signal intensity at different frequencies may contribute to its nonlinear behavior (Sun et al., 2023).

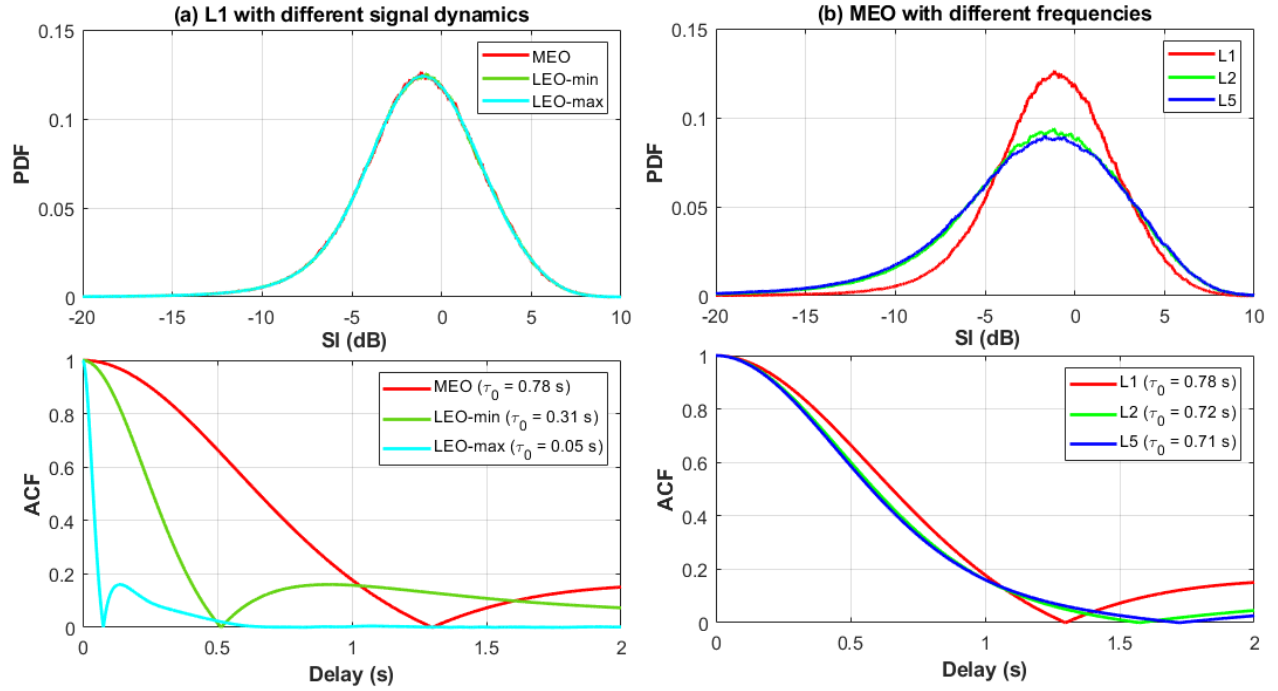


FIGURE 3 PDF (top) and ACF (bottom) of the signal intensity simulated for (a) different signal dynamics with L1 frequency and (b) different frequencies with the MEO case.

4 STATISTICAL ANALYSIS ON SCINTILLATION SIGNAL CHARACTERISTICS

4.1 Scintillation Signal Characteristics

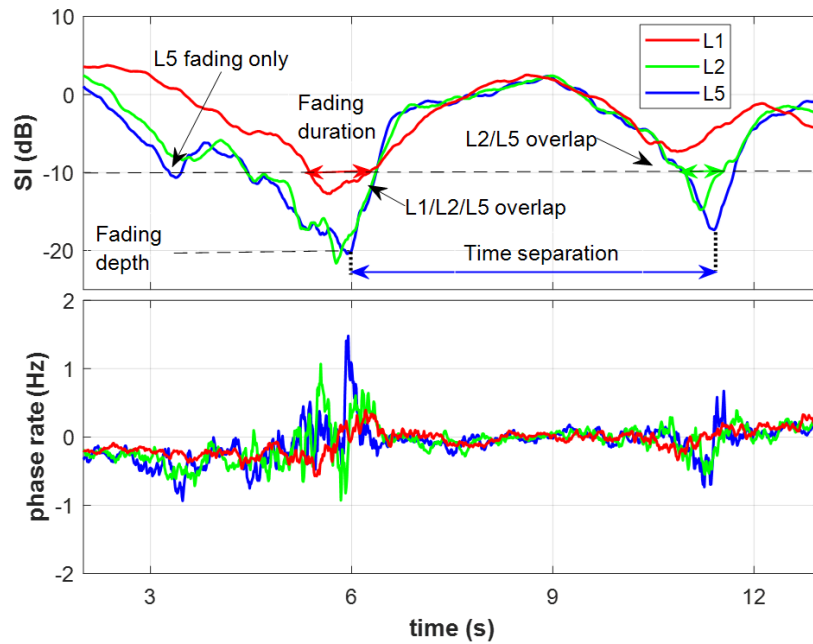


FIGURE 4 Illustration of signal fading characteristics with a -10 dB threshold and phase rate.

In this study, we analyzed the characteristics of signal fading and the phase rate disturbances. We applied a -10 dB threshold to detect fading events within the simulated signal intensity. Figure 4 illustrates the signal fading in L1, L2, and L5 with a -10 dB threshold, providing definitions of key characteristics, including the time separation between fades, fading duration, fading depth, and fading overlaps across different frequency signals (Jiao et al., 2016). We further calculated statistics of the phase rate specifically within the -10 dB signal fading events (Jiao et al., 2016).

4.2 Statistical Analysis of Scintillation Characteristics

4.2.1 Time Separation

The time separation between signal fades is a critical factor that indicates how frequent signal fades occur, potentially leading to cycle slips or loss of signal lock during the tracking process. This characteristic is essential for modeling the stochastic process of deep fades affecting signal reacquisition (Jiao et al., 2016; Sun et al., 2021). Fading time separation is defined as the time difference between the peak points of consecutive fades, as illustrated in Figure 4. This feature directly corresponds to the number of fades occurring per unit of time, as indicated in the legends in Figure 5.

Figure 5 shows histograms of the time separation between fades below -10 dB for L1, L2, and L5 signals. Subplots for MEO, LEO-min, and LEO-max cases depict the distribution of time separations for different signal dynamics. The results indicate that higher signal dynamics lead to more frequent fades, consistent with the prior analysis in Morton et al. (2020). The average time separation between fades for L1 frequency in the MEO case is 21.69 s whereas it is 8.55 s and 1.30 s in the LEO-min and LEO-max cases, respectively. This observation is consistent with the proportional relationship between the scintillation time scale, τ_0 , and the ρ_F/v_{eff} values across various signal dynamics, as presented in Figure 3a.

The results reveal considerably shorter time separations between fades for lower frequencies (L2 or L5) compared to L1. For instance, the L2 and L5 signals exhibit an average time separation of 0.39 s and 0.34 s in the LEO-max case, while the L1 signal experiences an average time separation of 1.30 s. These extremely frequent fades in lower frequency signals with higher dynamics potentially lead to a higher occurrence of cycle slips or even prolonged signal loss during reacquisition.

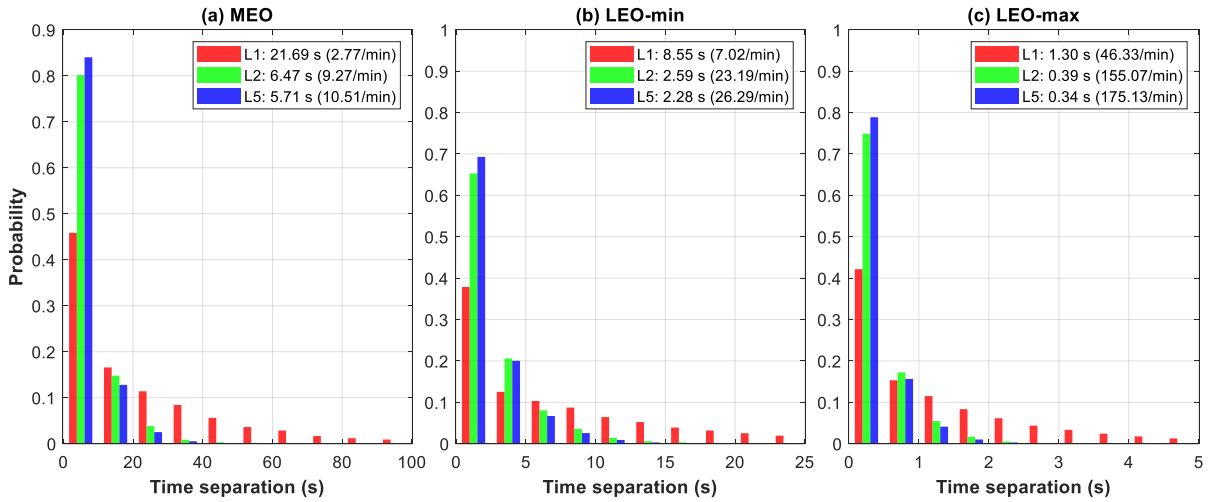


FIGURE 5 Normalized probability histograms of time separation between fades below -10 dB for L1, L2, and L5 frequencies in signal dynamics of (a) MEO, (b) LEO-min, and (c) LEO-max cases. The legends indicate the average time separation and the average number of fades per minute across simulation cases.

Figure 6 illustrates much significant variations in time separations for signals at different frequency bands far from the L-band, including VHF, UHF, and S-band, under varying signal dynamics: MEO, LEO-min, and LEO-max. For example, the S-band signal exhibits very few fades below -10 dB, with an average time separation of 112.59 s, even under high signal dynamics in the LEO-max case. In contrast, the VHF signal experiences extremely frequent fades, with an average time separation of 0.03 s in the LEO-

max case. This implies that deep fades occur at a rate of 2320.71 times per minute in VHF signals, which can drastically degrade the tracking process of such signals under high signal dynamics.

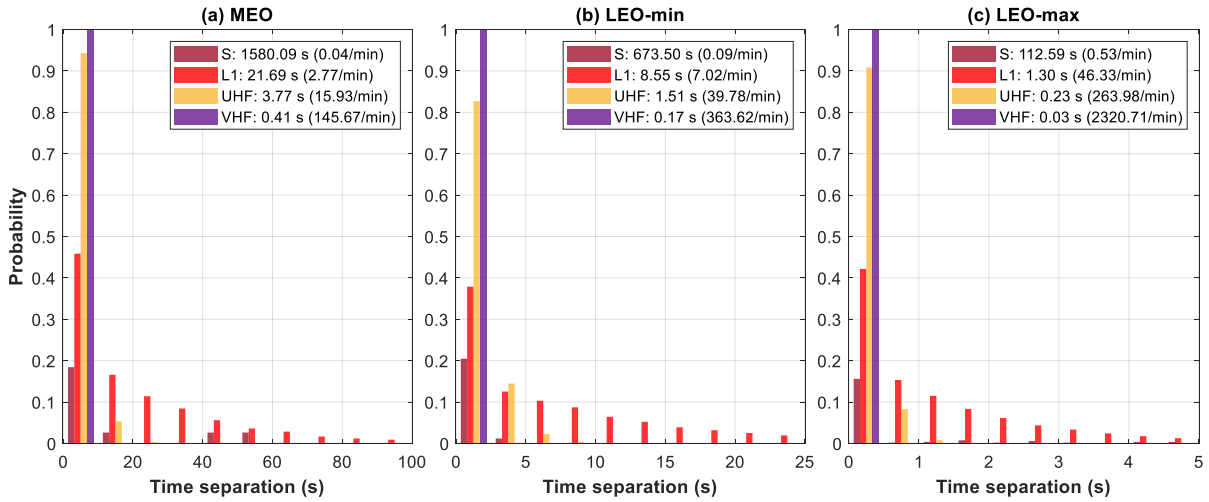


FIGURE 6 Normalized probability histograms of time separation between fades below -10-dB for VHF, UHF, L1, and S-band frequencies in signal dynamics of (a) MEO, (b) LEO-min, and (c) LEO-max cases. The legends indicate the average time separation and the average number of fades per minute across simulation cases.

4.2.2 Fading Duration

Fading duration is the time interval during which the signal intensity remains below the threshold. Longer fading durations increase the risk of cycle slips or loss of signal lock in receiver tracking. A receiver can reacquire a signal once the intensity recovers above the tracking threshold; therefore, fading duration is a crucial parameter determining the reacquisition capability of receivers (Jiao et al., 2016; Sun et al., 2021).

Figure 7 shows histograms of fading durations for signals at L1, L2, and L5 frequencies across various signal dynamics. The results indicate that the LEO-max case with higher signal dynamics exhibits much shorter fading durations, with an average value of 0.020 s, compared to the MEO case, where the average fading duration is 0.319 s for the L1 signal. Similar to the observations in the time separation between fades, this finding aligns well with the proportional relationship between the scintillation time scale, τ_0 , and the p_F/v_{eff} values across different signal dynamics, as presented in Figure 3a.

Each subplot of Figure 7 further reveals that lower frequencies (L2 or L5) tend to have slightly longer signal fades. In the MEO case, the average fading duration for L1, L2, and L5 is 0.319 s, 0.363 s, and 0.364 s, respectively. This finding aligns with the theory suggesting that lower frequency signals are more susceptible to scintillation (Jiao et al., 2016) as there will be more frequent deeper fades at lower frequencies, resulting in a higher occurrence of longer fades for a given threshold, -10 dB in this study.

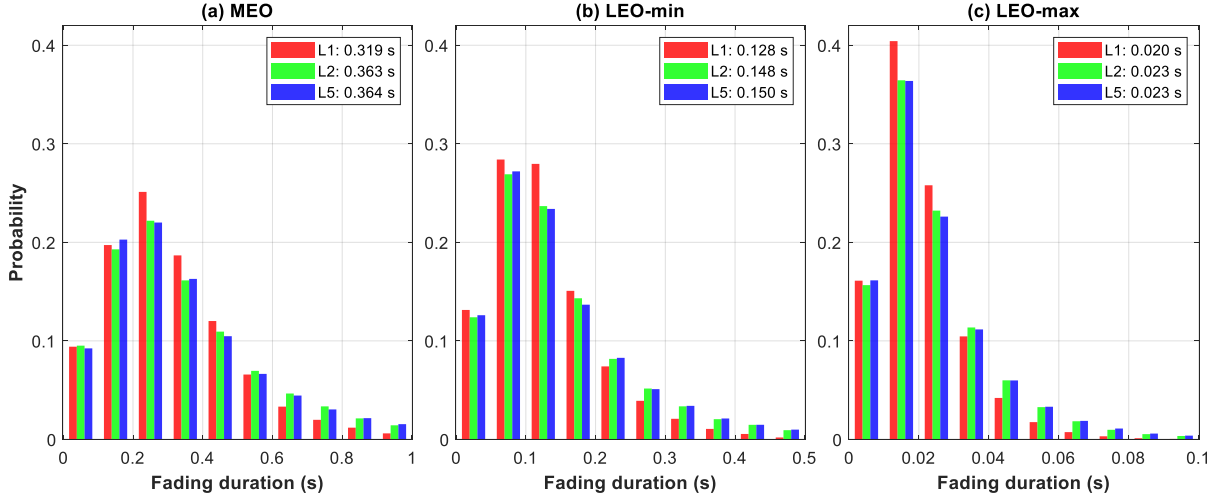


FIGURE 7 Normalized probability histograms of fading duration below -10 dB for L1, L2, and L5 frequencies in signal dynamics of (a) MEO, (b) LEO-min, and (c) LEO-max cases. The legends indicate the average fading duration across simulation cases.

Figure 8 illustrates similar trends in the distributions of fading durations for signals at frequencies in VHF, UHF, L1, and S-band across various signal dynamics scenarios. The results emphasize the dependence of fading duration on signal dynamics for all frequency bands. In each signal dynamics scenario, the average fading duration increases as frequency decreases down to UHF, likely due to the more frequent occurrence of deeper fades with higher scintillation levels at lower frequencies. However, VHF signals deviate from this trend, exhibiting exceptionally shorter fades attributed to extremely strong scintillations at the lowest frequency band. This deviation could stem from violations in fundamental assumptions made in the simulation model, originally designed for L-band signals. Therefore, further updates on the simulation model would be necessary to accurately simulate scintillation effects across a broader frequency spectrum.

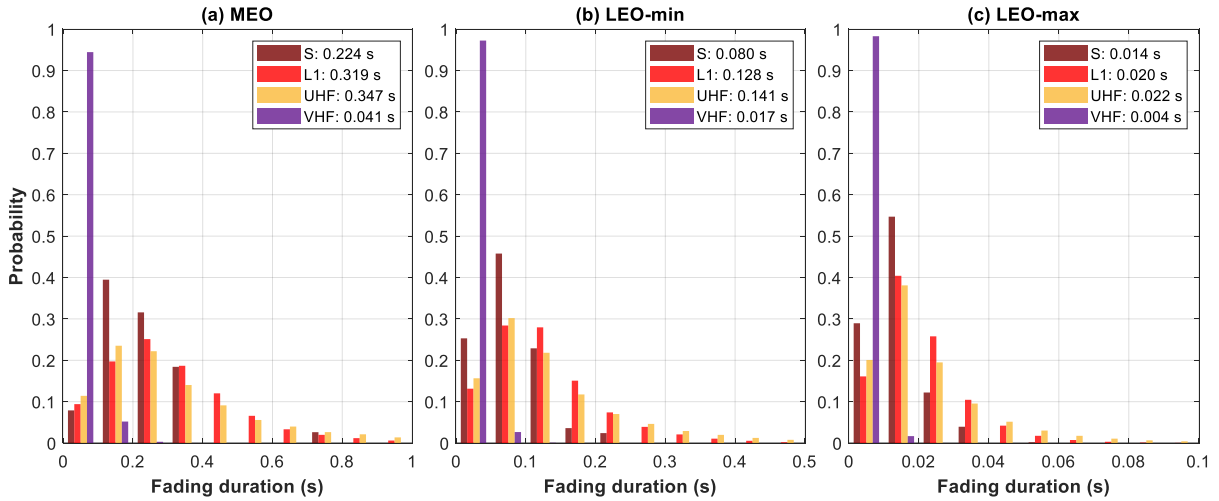


FIGURE 8 Normalized probability histograms of fading duration below -10 dB for VHF, UHF, L1, and S-band frequencies in signal dynamics of (a) MEO, (b) LEO-min, and (c) LEO-max cases. The legends indicate the average fading duration across simulation cases.

4.2.3 Fading Depth

The fading depth is defined as the peak signal intensity within the -10 dB fading events, as illustrated in Figure 4. Figure 9 shows histograms of fading depth below -10 dB for signals at L1, L2, and L5 frequencies across different signal dynamics scenarios:

MEO, LEO-min, and LEO-max. The results reveal that the distribution of fading depth remains consistent across all signal dynamics scenarios. This suggests that fading depth exhibits less significant dependence on signal dynamics. However, it is noteworthy that higher signal dynamics can lead to more frequent fades within a specific time interval.

Furthermore, the results in Figure 9 show that lower frequencies (L2 and L5) have a notably higher probability of deeper fading compared to L1 for all signal dynamics scenarios. This result can be attributed to the stronger scintillation levels resulting from higher scattering strength (U) at lower frequencies, leading to larger variations in signal intensity and consequently, a higher occurrence of deeper fades at lower frequencies.

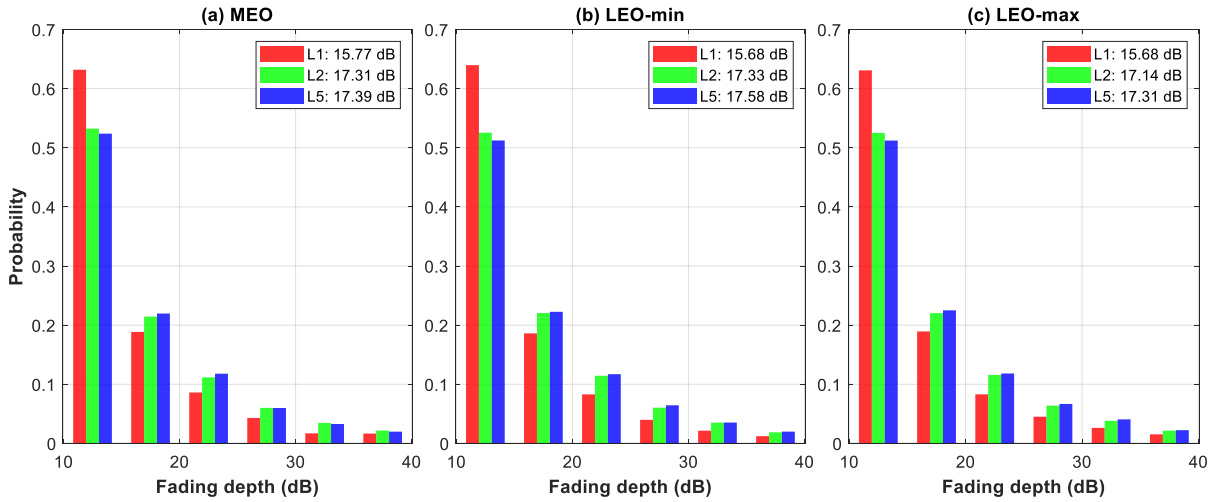


FIGURE 9 Normalized probability histograms of fading depth below -10 dB for L1, L2, and L5 frequencies in signal dynamics of (a) MEO, (b) LEO-min, and (c) LEO-max cases. The legends indicate the average fading depth across simulation cases.

Figure 10 presents histograms of fading depth for signals at frequencies in VHF, UHF, GPS L1, and S-band across signal dynamics scenarios. The results across frequencies at different bands in Figure 10 show consistent distributions of fading depth across various signal dynamics scenarios, aligning with the L-band results in Figure 9. These findings further highlight the contribution of frequency to the distribution of fading depth. It is noteworthy that the relatively lower average values of fading depth for the LEO-max case can be attributed to extremely short fades, potentially missing peak values within a given sampling interval.

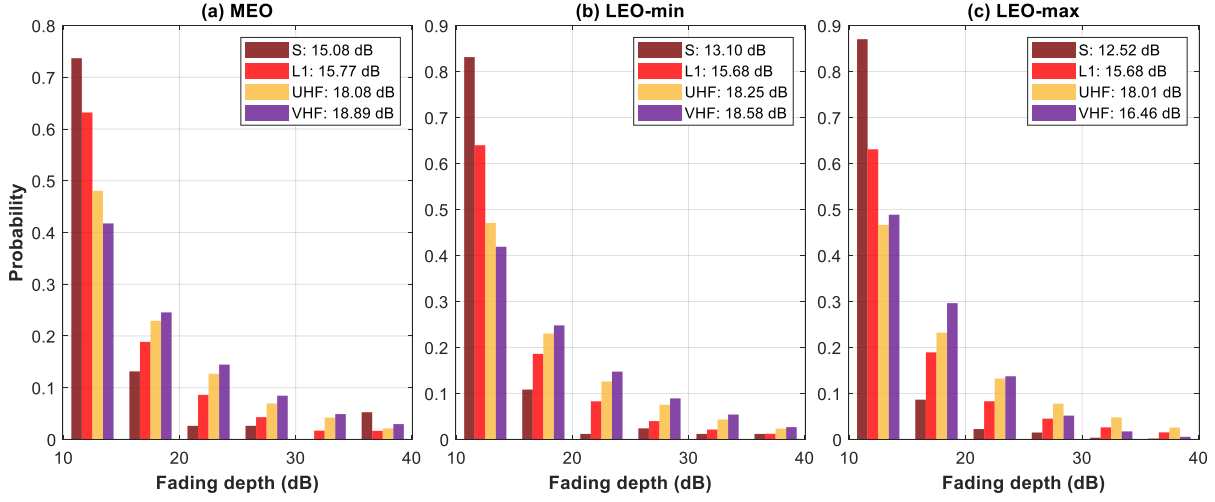


FIGURE 10 Normalized probability histograms of fading depth below -10 dB for VHF, UHF, L1, and S-band frequencies in signal dynamics of (a) MEO, (b) LEO-min, and (c) LEO-max cases. The legends indicate the average fading depth across simulation cases.

4.2.4 Phase Rate

Figure 11 presents the distribution of phase rates during fading with a -10 dB threshold for signals at L1, L2, and L5 frequencies across various signal dynamics scenarios: MEO, LEO-min, and LEO-max. The LEO-max case exhibits significantly larger phase rates compared to the MEO case, as discussed in Morton et al. (2022). A slight increase in phase rates is also observed at lower frequencies.

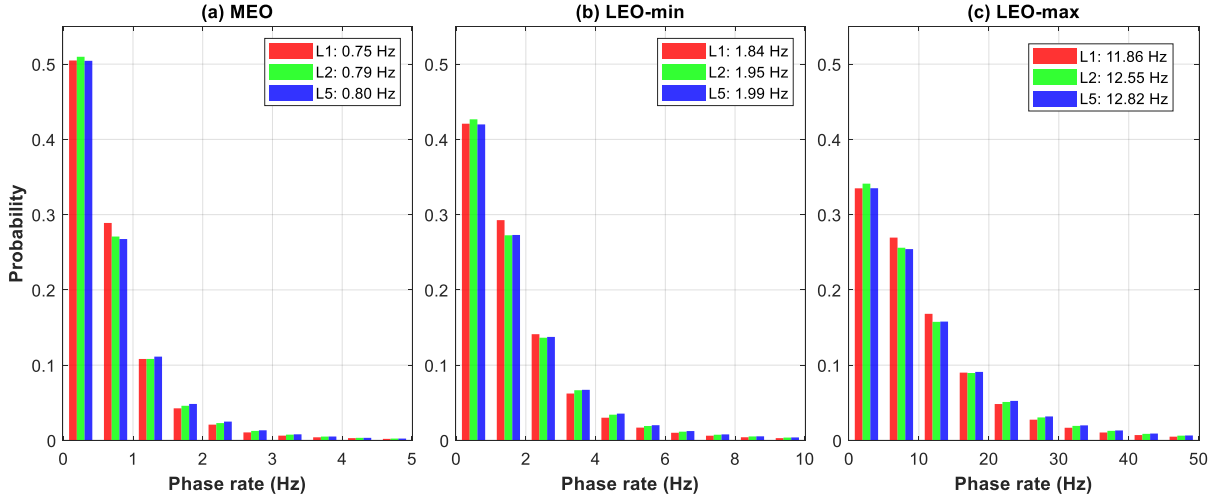


FIGURE 11 Normalized probability histograms of phase rate within -10 dB fading events for L1, L2, and L5 frequencies in signal dynamics of (a) MEO, (b) LEO-min, and (c) LEO-max. The legends indicate the average phase rate across simulation cases.

Figure 12 presents the distribution of phase rates during fades below -10 dB for signals at frequencies in VHF, UHF, L1, and S-band signals across signal dynamics scenarios. The results demonstrate a pronounced increase in phase rate at lower frequency bands. The multi-frequency results across various signal dynamics show a nearly inverse proportionality of phase rate with the ρ_F/v_{eff} values for a given frequency, where higher phase rates are exhibited at lower ρ_F/v_{eff} values under higher signal dynamics.

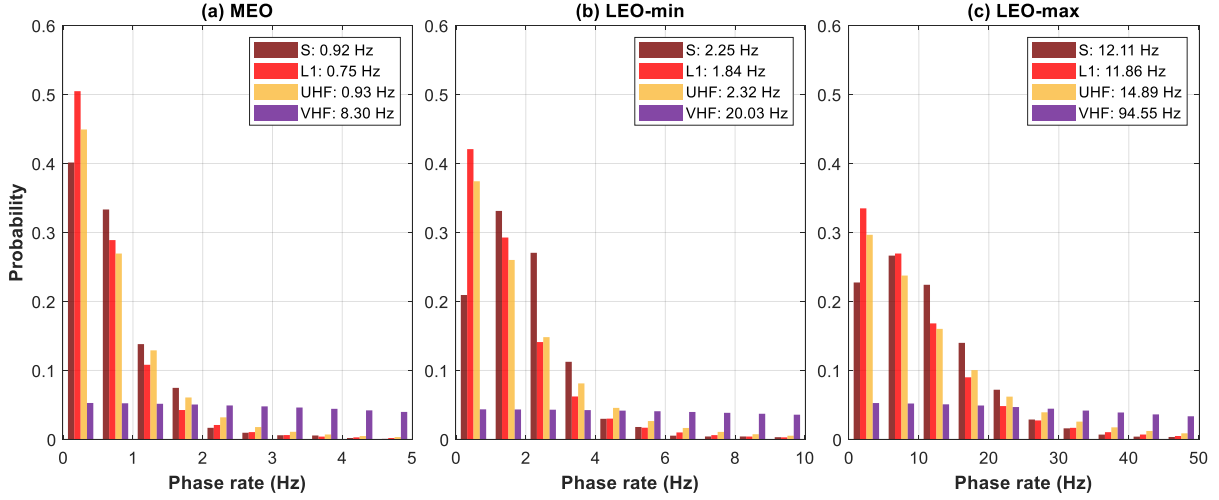


FIGURE 12 Normalized probability histograms of phase rate within -10 dB fading events for VHF, UHF, L1, and S-band frequencies in signal dynamics of (a) MEO, (b) LEO-min, and (c) LEO-max. The legends indicate the average phase rate across simulation cases.

4.2.5 Analysis of Temporal Characteristics in Scintillation across Signal Dynamics

To quantitatively analyze the relationship between temporal characteristics in scintillation and different signal dynamics, we calculated the ratio of mean values for time separation between fades, fading duration, and phase rate across MEO, LEO-min, and LEO-max cases with different ρ_F/v_{eff} values. Figure 13 illustrates the ratio of mean values for time separation between fades and fading duration of LEO-min (or LEO-max) over MEO for all simulated frequencies. Additionally, we calculated the ratio of mean values for phase rate of MEO over LEO-min (LEO-max), considering that the temporal scale is inversely proportional to phase rate. The black dash lines represent the ratio of ρ_F/v_{eff} for LEO-min (or LEO-max) over MEO.

The results consistently reveal that the ratio of mean values for fading separation and duration of LEO-min (or LEO-max) over MEO closely aligns with the ratio of ρ_F/v_{eff} of LEO-min (or LEO-max) over MEO across all simulated frequencies. Regarding the phase rate, the ratio of mean values of MEO over LEO-min (or LEO-max) corresponds to the ratio of ρ_F/v_{eff} of LEO-min (or LEO-max) over MEO.

Deviation in the ratio of mean values is observed at the lowest frequency in the VHF and the highest frequency in the S-band. For the VHF signal, excessively rapid variations in signal intensity under LEO signal dynamics may lead to shorter signal fades for a fixed sampling interval, potentially causing an overestimation of the temporal scales in signal fading. For the S-band signal, the limited number of signal fades during the simulation period may result in a lack of statistical significance in the mean values of each characteristic.

The results across various frequencies indicate that the temporal scale of signal fading (fading separation and duration) is directly proportional to ρ_F/v_{eff} , consistent with the findings in Xu et al. (2022) where the decorrelation time (τ_0), the representative temporal scale of scintillation, is proportional to ρ_F/v_{eff} as discussed in Section 3.3. Moreover, this suggests that we can accurately estimate the temporal characteristics of signal fading at two different signal dynamics by scaling them with the ratio of their ρ_F/v_{eff} values.

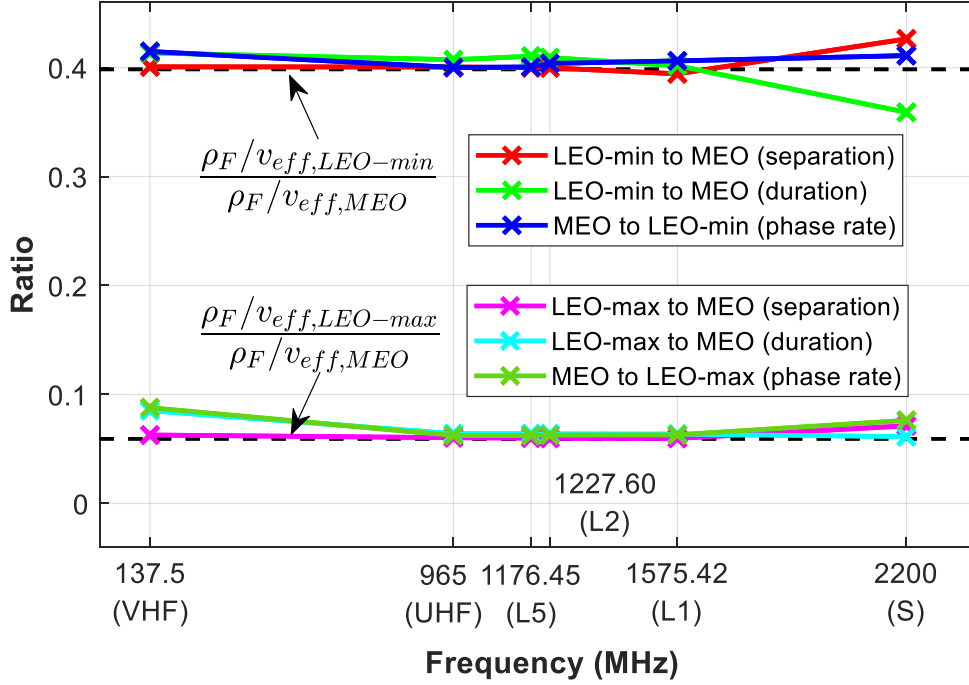


FIGURE 13 Ratio of mean values for fading time separation, duration, and phase rate for LEO-min (or LEO-max) over MEO across simulated frequencies: VHF, UHF, L5, L2, L1, and S-band.

4.2.6 Fading Overlaps

The simultaneous occurrence of signal fading across different frequencies is crucial because it can directly lead to the loss of satellite tracking when the satellite transmits signals at multiple frequencies. Sufficient time separations in concurrent fades allow us to leverage frequency diversity during deep fading for inter-frequency aiding, as observed in Jiao et al. (2016). As illustrated in Figure 4, fading overlaps are instances where the time durations of fading events overlap among different signals. Table 3 summarizes the time fractions of all combinations of fading overlaps in L1, L2, and L5 signals for each frequency over a simulation period of 1,000 min across signal dynamics scenarios: MEO, LEO-min, and LEO-max.

The results show that concurrent fading on all three frequencies is highly improbable, with fading overlap percentages less than 13.58 % for L1, 3.53 % for L2 and 3.10 % for L5 signals across all signal dynamics scenarios. Even concurrent fading in two frequencies, such as between L1 and L2 or L1 and L5, occurs at low percentages, less than 5.08 % or 3.76 %, respectively. These observations align with the findings of real scintillation observations presented in Jiao et al. (2016), highlighting the rarity of concurrent fading in the nearby L-band frequencies, L1, L2, and L5.

An interesting observation regarding fading overlaps is that the percentages of each type of fading remain almost consistent across different signal dynamics scenarios. The consistent fading overlaps across signal dynamics may be attributed to the proportional time scale of fading characteristics with ρ_F/v_{eff} values, as discussed in Section 4.2.5. This suggests that a consistent proportion of time separation among fades at different frequencies effectively contributes to frequency diversity in frequency-aiding tracking, even in cases of high signal dynamics. Slight increases in fading overlaps at higher signal dynamics may be attributed to the discretization of samples, particularly with shorter fades occurring under higher signal dynamics.

TABLET 3

The Percentage of Signal Fading Overlaps in Simulated L1, L2, and L5 Signals.

Fading band	Scenarios	L1 only	L2C only	L5 only	Concurrent L1/L2	Concurrent L1/L5	Concurrent L2/L5	Concurrent L1/L2/L5
L1	MEO	79.76%	/	/	17.18%	14.65%	/	11.60%
	LEO-min	77.80%	/	/	18.98%	16.20%	/	12.98%
	LEO-max	77.56%	/	/	19.56%	16.46%	/	13.58%
L2	MEO	/	38.18%	/	4.50%	/	60.36%	3.03%
	LEO-min	/	36.77%	/	4.95%	/	61.67%	3.38%
	LEO-max	/	37.26%	/	5.08%	/	61.19%	3.53%
L5	MEO	/	/	46.29%	/	3.37%	53.01%	2.67%
	LEO-min	/	/	45.28%	/	3.70%	53.99%	2.96%
	LEO-max	/	/	45.52%	/	3.76%	53.82%	3.10%

5 CONCLUSIONS

This paper extends simulation studies presented in Morton et al. (2022) to investigate scintillation effects on signals transmitted from LEO satellites across various frequency bands. Our approach involves using a physics-based scintillation model developed by Xu et al. (2020), which provides a comprehensive representation of scintillation with distinct signal dynamics by employing the time-to-space parameter, ρ_F/v_{eff} . The model incorporates frequency scaling of modeling parameters from initialization data, enabling it to effectively represent scintillation for a specific frequency assuming identical irregular structures.

By utilizing this simulator, we conducted scintillation simulations under varying signal dynamics, including MEO, LEO-min, and LEO-max cases, across different frequencies at diverse frequency bands including VHF, UHF, L (L1, L2, and L5) and S-band. It is important to note that some of the fundamental assumptions made in the scintillation simulation model may not be applicable to lower frequency bands, such as VHF signals, where the model only assumes forward scattering. Additional research is underway to update the model to accommodate lower frequencies.

The simulation results reveal more frequent and shorter fades with a higher phase rate under increased signal dynamics, consistent with the findings in Morton et al. (2022). Additionally, the new findings across various frequency bands indicate that signals at lower frequencies exhibit more frequent fades and greater phase dynamics. Analyses across various frequency bands provide quantitative evidence that distinct signal dynamics determine the temporal scales of scintillation and associated signal fading features, including time separation between fades and fading duration, both proportional to ρ_F/v_{eff} , and phase rate, inversely proportional to ρ_F/v_{eff} .

Our simulation results also show consistently low fading overlap percentages across L1, L2, and L5 frequencies under identical phase screen sequence, across various scenarios of signal dynamics. This phenomenon aligns well with observations from ground receivers of multi-frequency GNSS signals, as reported by Jiao et al. (2016). This consistently low fading overlap pattern highlights the potential for leveraging frequency diversity to improve tracking and navigation operations (Yang et al., 2020), even within dynamic platforms like LEO satellite transmissions. These findings contribute to our understanding of ionospheric scintillation effects and their impact on LEO satellite-based communication and navigation.

REFERENCES

Enge, P., Ferrell, B., Bennett, J., Whelan, D., Gutt, G., Lawrence, D., "Orbital Diversity for Satellite Navigation," Proceedings of the 25th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS 2012), Nashville, TN, September 2012, pp. 3834-3846.

- Garrison, J. L., Nold, B., Lin, Y. -C., Pignotti, G., Piepmeier, J. R., Vega, M., Fritts, M., DuToit, C., & Knuble, J. (2017). Recent results on soil moisture remote sensing using P-band signals of opportunity, 2017 International Conference on Electromagnetics in Advanced Applications (ICEAA), Verona, Italy, 1604-1607. <https://doi.org/10.1109/ICEAA.2017.8065595>
- Hsiao, T. Y., Huang, C. -y., Yeh, W. -H., Tseng, T. -P., Chen, K. -L., Macalalad, E. P., Vallar, E. A., & Galvez, M. C. D. (2022). The observations of localize ionospheric scintillation structure by FORMOSAT-7/COSMIC-2 Tri-band Beacon network. *Terrestrial, Atmospheric and Oceanic Sciences*, 33(10). <https://doi.org/10.1007/s44195-022-00010-6>
- Jiao, Y., Xu, D., Morton, Y., & Rino, C. (2016). Equatorial Scintillation Amplitude Fading Characteristics Across the GPS Frequency Bands. *NAVIGATION*, 63(3), 267-281. <https://doi.org/10.1002/navi.146>
- Jiao, Y., Xu, D., Rino, C. L., Morton, Y. T., & Carrano, C. S. (2018). A Multifrequency GPS Signal Strong Equatorial Ionospheric Scintillation Simulator: Algorithm, Performance, and Characterization. *IEEE Transactions on Aerospace and Electronic Systems*, vol. 54, no. 4, pp. 1947-1965. <https://doi.org/10.1109/TAES.2018.2805232>
- Joerger, M., Gratton, L., Pervan, B., & Cohen, C. E. (2010). Analysis of Iridium-Augmented GPS for Floating Carrier Phase Positioning. *NAVIGATION*, 57(2), 137-160. <https://doi.org/10.1002/j.2161-4296.2010.tb01773.x>
- Kassas, Z. M. (2020). Navigation from Low-Earth Orbit. In Y. T. J. Morton, F. Diggelen, J. J. Spilker, B. W. Parkinson, S. Lo & G. Gao (Eds.), *Position, Navigation, and Timing Technologies in the 21st Century* (pp. 1381-1412). IEEE. <https://doi.org/10.1002/9781119458555.ch43b>
- Morton, Y. J., Xu, D., Jiao, U. (2022). Ionospheric Scintillation Effects on Signals Transmitted from LEO Satellites. *Proceedings of the 35th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2022)*, Denver, CO, 2980-2988. <https://doi.org/10.33012/2022.18341>
- Neinavaie, M., Khalife, J., & Kassas, Z. M. (2021). Exploiting Starlink Signals for Navigation: First Results. *Proceedings of the 34th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2021)*, St. Louis, MS, 2766-2773. <https://doi.org/10.33012/2021.18122>
- NOAA (2009). NOAA Table of Polar-orbiting Satellites Equator Crossing Times and Frequencies (CGMS-37, NOAA-WP-37). Retrieved from <https://noaasis.noaa.gov/NOAASIS/pubs/spacecraft%20effects.pdf>
- Prol, F. S., Morales Ferre, R., Saleem, Z., Välisuo, P., Pinell, C., Lohan, E.-S., Elsanhoury, M., Elmusrati, M., Islam, S., Çelikbilek, K., Selvan, K., Yliaho, J., Rutledge, K., Ojala, A., Ferranti, L., Praks, J., Zahidul H. Bhuiyan, M., Kaasalainen, S., & Kuusniemi, H. (2022). Position, Navigation, and Timing (PNT) Through Low Earth Orbit (LEO) Satellites: A Survey on Current Status, Challenges, and Opportunities. *IEEE Access*, 10, 83971-84002. <https://doi.org/10.1109/ACCESS.2022.3194050>
- Reid, T. G. R., Walter, T., Enge, P. K., Lawrence, D., Cobb, H. S., Gutt, G., O'Conner, M. (2020). Navigation from Low-Earth Orbit. In Y. T. J. Morton, F. Diggelen, J. J. Spilker, B. W. Parkinson, S. Lo & G. Gao (Eds.), *Position, Navigation, and Timing Technologies in the 21st Century* (pp. 1359-1379). IEEE. <https://doi.org/10.1002/9781119458555.ch43a>
- Rino, C., Breitsch, B., Morton, Y., Jiao, Y., Xu, D., & Carrano, C. (2018). A compact multi-frequency GNSS scintillation model. *NAVIGATION*, 65(4), 563-569. <https://doi.org/10.1002/navi.263>
- Sun, A. K., Chang, H., Pullen, S., Kil, H., Seo, J., Morton, Y. J., & Lee, J. (2021). Markov chain-based stochastic modeling of deep signal fading: Availability assessment of dual-frequency GNSS-based aviation under ionospheric scintillation. *Space Weather*, 19, e2020SW002655. <https://doi.org/10.1029/2020SW002655>
- Sun, A. K., Pi, X., Rino, C., & Lee, J. (2023). Statistical uncertainty in the frequency dependence of the intensity scintillation index (S4). *Radio Science*, 58, e2023RS007659. <https://doi.org/10.1029/2023RS007659>

Xu, D., Morton, Y. T. J., Rino, C. L., Carrano, C. S., & Jiao, Y. (2020). A two-parameter multifrequency GPS signal simulator for strong equatorial ionospheric scintillation: modeling and parameter characterization. *NAVIGATION*. 67(1), 181-195. <https://doi.org/10.1002/navi.350>

Xu, J., Morton, Y. J., Xu, D., Jiao, Y., & Hinks, J. (2023). Tracking GPS-like Signals Transmitted from LEO Satellites and Propagated Through Ionospheric Plasma Structures. *Proceedings of the 36th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2023)*, Denver, CO, 2813-2822.

Yang, R., Xu, D., & Morton, Y. T. (2020). Generalized multi-frequency GPS carrier tracking architecture: design and performance analysis. *IEEE Transactions on Aerospace and Electronic Systems*, 56(4), 2548-2563. <https://doi.org/10.1109/TAES.2019.2948535>